

How models learn

Machine learning replaces some hand-written decision rules with parameters adjusted from examples.

15–20 MINUTES

Prerequisite: visual features

INTERACTIVE SCORE

1 • Example

An input paired with supervision.

2 • Model

A parameterized mapping.

3 • Training

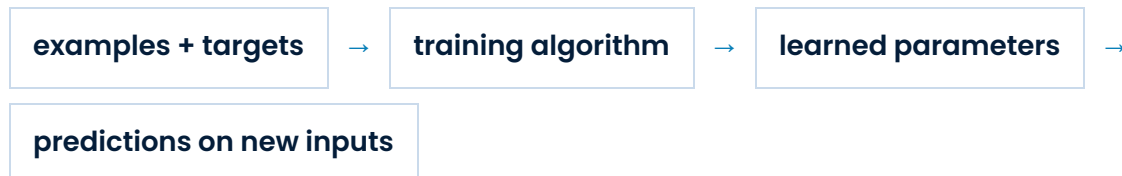
Adjust parameters using error.

4 • Inference

Apply frozen parameters.

Rules, parameters, and examples

In ordinary programming, a developer writes rules and the program applies them to data. In supervised machine learning, the developer defines a model family and learning procedure, supplies labeled examples, and training selects parameter values that make useful predictions.



An **example** is one input, such as an image. A **target** is the desired answer supplied during training. A **parameter** is a numeric value the learning algorithm may change. A **hyperparameter**, such as learning rate or number of epochs, controls training and is chosen outside the learned parameter set.

Central mental model

Training searches for parameters using labeled examples. Inference uses the resulting parameters to predict without seeing the target.

A tiny model

Suppose a deliberately simple classifier uses a single measured feature: how red an image region is. It calculates $\text{score} = \text{weight} \times \text{redness} + \text{bias}$, then maps the score to a probability. Training changes the weight and bias. This model can learn that higher redness supports “red cube,” but it cannot distinguish a red ball because its input feature contains no shape evidence.

Interactive score

Model score before probability: **0**



Probability after sigmoid: **0.500**

The sigmoid converts any real score to a value between 0 and 1. It does not make the value true; it is still a model prediction.

Supervision for detection

For image classification, a target can be one class. For object detection, supervision includes each object's class and bounding box. The training procedure compares several kinds of prediction with these targets, so “the loss” is often a combination of terms.

Labels describe the annotation policy, not nature itself. If partly hidden cubes are inconsistently labeled, the model receives inconsistent supervision. Better algorithms cannot recover information the targets fail to specify clearly.

Applied to our cubes

REPOSITORY OBSERVATION The source dataset contains images paired with YOLO detection labels. Each usable row identifies a class and normalized box. Training began from pretrained weights and produced `best.pt`.

IMPORTANT DISTINCTION `best.pt` stores learned parameters and related checkpoint information. It is the result of a training process, not the training data or the learning algorithm itself.

Project anchor: model artifact notes.

Check your understanding

What happens during inference?

- ☐ Labels update model parameters
- ☐ Fixed parameters produce predictions
- ☐ Test examples choose hyperparameters

Explain it back

Use the words example, target, parameter, training, and inference to explain a learned cube detector. Then say why a score is not a guarantee.

Primary sources

- Deep Learning, Chapter 5 — machine-learning basics.
- PyTorch Learn the Basics — executable training workflow.